NANDHAKUMAR G

Mechanical Engineering Undergraduate (2027 Batch)

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Analytical and detail-oriented Mechanical Engineering student with hands-on industrial traineeship experience in raw material systems and continuous casting operations, combined with professional 3D CAD modeling certifications. Passionate about reading and interpreting mechanical drawings, technical documentation, and electromechanical systems integration. Proven record of developing structured technical specs, optimizing part design parameters, and executing high-reliability systems engineering projects.

IMPACT SNAPSHOT & KEY MILESTONES

8	3	4
CERTIFICATIONS	INDUSTRY INTERNSHIPS	ENGINEERING PROJECTS

EXECUTIVE FOCUS

- ✓ **Technical Documentation:** Experienced in creating and analyzing engineering reports, parts lists, and detailed technical specifications to bridge the gap between design and physical assembly operations.
- ✓ **Drawing Interpretation:** Certified and proficient in interpreting mechanical engineering drawings, electrical schematics, assembly layouts, and structural tolerance dimensions.
- ✓ **Systems Engineering:** Skilled in integrating controls firmware with sensor networks and hardware actuators, maintaining a high level of academic excellence and attention to parts diagnostics.

CORE COMPETENCIES

ENGINEERING SUPPORT & TECH DOC

Drawing Interpretation · Technical Writing · Parts Specification · Diagnostics Support

MECHANICAL DESIGN & CAD

Parametric Assemblies · Tolerance Stacks · DFM Guidelines · FEA Simulation

TOOLS & METHODS

Fusion 360	SolidWorks	Siemens NX	AutoCAD
Python	C++	MATLAB	COMSOL

PROFESSIONAL ATTRIBUTES

Analytical Problem Solving:
Capable of debugging electromechanical control loops and structural mechanical clearances using multi-faceted diagnostic approaches.

Cross-Functional Communication:
Articulates core technical principles and system layouts to interdisciplinary teams, ensuring accurate specifications transition from draft to production.

Rapid Learning & Growth Mindset:
Consistently acquires certifications and applies engineering concepts to core operations, demonstrating a quick learning curve for mechanical assemblies.

PROFESSIONAL EXPERIENCE (INTERNSHIPS)

Reverse Engineering Intern
Femlogic Technologies Pvt. Ltd., Chennai — June 2026

- Performed dimensional inspection and geometric data capture of legacy mechanical parts using precision metrology tools (calipers, micrometers, and thread gauges).
- Reconstructed physical components into fully editable parametric 3D assemblies, resolving design deviations and model geometry errors.
- Validated clearance configurations, fit parameters, and tolerance stacks to ensure compatibility with modern mechanical assemblies.
- Modified CAD files to optimize structures for additive manufacturing (3D printing), optimizing print orientation, tolerances, and support layouts.

Industrial Engineering Trainee
Steel Authority of India Ltd. (SAIL), Salem Steel Plant — December 2025

- Inspected heavy steel rolling mill structures, power transmission machinery drives, and continuous casting systems.
- Analyzed and interpreted plant mechanical engineering drawings, flow diagrams, and hydraulic schematic charts.
- Documented maintenance protocols, equipment operational checklists, and safety manuals for plant machinery.
- Collaborated with senior maintenance technicians to analyze equipment performance and trace components wear histories.

Product Design Intern
Femlogic Technologies Pvt. Ltd., Chennai — June 2025 - July 2025

- Assisted in parametric 3D CAD modeling and design assembly validation of Vande Bharat seating structural modules.
- Reviewed and updated structural interface specifications, ensuring compliance with manufacturing (DFM) requirements.
- Drafted comprehensive design validation reports and parts documents to support cross-functional verification teams.

SELECTED PROJECT PORTFOLIO

Project EcoFloat | Niral Hackathon
LEAD DEVELOPER & DESIGNER | ROS · IoT · Fusion 360

- Spearheaded development of a dual-function autonomous robotic boat engineered to simultaneously remove physical aquatic trash and detect chemical runoff.
- Implemented a real-time toxicity sensing framework powered by IoT and ROS, designed to transition marine cleanup from reactive manual labor to a proactive, data-driven automated system.
- Architected the platform's physical design in Fusion 360, prioritizing solar-powered endurance and scalability for deployment by municipal and port authorities.

Regenerative Braking System for Electric Vehicle
LEAD RESEARCH & DEVELOPER | MATLAB Simulink · BMS

- Designed and developed a regenerative braking system to improve energy efficiency in an electric vehicle prototype.
- Converted kinetic energy during braking into electrical energy, optimizing recovery using MATLAB Simulink models.
- Integrated motor-generators and battery management systems (BMS), ensuring smooth operation and increasing battery lifespan by 20%.

Advanced Aquatic Monitoring System
SYSTEMS ENGINEER | Hyperspectral Imaging · Sensor Fusion

- Designed and developed an autonomous floating platform equipped with a synthesized high-resolution hyperspectrometer to conduct real-time environmental analysis of water bodies.
- Processed hyperspectral imaging data to accurately detect and quantify microplastic concentrations, while simultaneously mapping chlorophyll levels to assess aquatic health and potential algal blooms.
- Integrated advanced sensor data algorithms, improving ecological monitoring efficiency and providing precise, actionable data for marine conservation and water quality management.

Hemispherical CPV Solar Tracking System
MECHANICAL DESIGN ENGINEER | COMSOL Multiphysics · Solar Vector Math

- Developed a hemispherical CPV solar tracking system with dual-axis tracking and adaptive weather control to improve energy efficiency.
- Designed hemispherical shell-shaped panels to enhance light absorption and angular coverage, validating the design through COMSOL Multiphysics® simulations.
- Optimized solar tracking calculations, making the system ideal for compact and wearable energy harvesting applications.

EDUCATION QUALIFICATIONS

Bachelor of Mechanical Engineering (2027 Batch) — 2023 - Present
St. Joseph's College of Engineering, Chennai, Tamil Nadu
Core Coursework: Thermodynamics, Fluid Mechanics, Machine Design, Control Systems

Higher Secondary Education (Class XII) — 2022 - 2023
Shri Swamy Matriculation Higher Secondary School, Salem, Tamil Nadu
Computer Science Stream

Secondary School Education (Class X) — 2020 - 2021
Shri Swamy Matriculation Higher Secondary School, Salem, Tamil Nadu
Matriculation State Board Curriculum

PROFESSIONAL CREDENTIALS

Autodesk Certified Fusion Designer
Professional CAD Design Credential

Certified Solidworks Designer
Mechanical Modeling & Validation Spec

Certified Siemens NX CAD Designer
Advanced Parametric Modeling Specialist

AutoCAD for 2D Modelling
Technical Drafting & Detailing Standard

Certified in Python Advanced
Computational Data Structures & Integration

Certified Frontend Developer
Core Interface Architect (HTML/CSS/JS)

Blender for 3D Modelling
Complex Surface Mesh Optimization

C++ Programming
Object-Oriented Software & Logic Design

LANGUAGES

English
Fluent

Tamil
Native